Draft sections for the PRD: warped embedding; lepton–meson realization

sections numbered after the PRD; equations from (101); bare numbers in brackets and equations below (78) refer to the PRD

15 Warped embedding and separation of the hard scale

This section replaces the flat common-tension disk of Sec. X by a confining warped background and asks under which condition the same-sign four-Ramond amplitude remains pointlike at $s \gg M_{\rm had}^2 \equiv 1/\alpha'_{\rm had}$.

15.1 Radial convolution

Take the Polchinski–Strassler (PS) background [1]

$$ds^2 = \frac{r^2}{R^2}\,\eta_{\mu\nu}dx^\mu dx^\nu + \frac{R^2}{r^2}\,dr^2 + \cdots\,, \qquad r \geq r_{\rm min}, \tag{101}$$

with the Ramond states as open strings on branes that extend along r. Local inertial momenta are $\tilde{p}_\mu = (R/r)\,p_\mu$, so $\alpha'\tilde{s} = \alpha'(R^2/r^2)\,s$. Strings localized near $r_{\rm min}$ have the four-dimensional Regge slope $\alpha'_{\rm had} = \alpha' R^2/r_{\rm min}^2$. Define

$$y = \frac{r_{\rm min}}{r} \in (0,1], \qquad z = \alpha'_{\rm had}\,s, \qquad \alpha'\tilde{s} = z\,y^2. \tag{102}$$

The rescaling $p \to \tilde{p}$ is uniform, so the local scattering angle equals the four-dimensional one and $x = (1-\cos\theta)/2$ is independent of r.

The four-dimensional amplitude is the radial integral of the local string amplitude weighted by the external profiles [PS, Eq. (7)]. For the same-stack sector the local amplitude is $A_{\rm SYM}(\tilde{p})\,V$; at fixed angle the r dependence of $A_{\rm SYM}(\tilde{p})$ is a power that PS absorb into the twist counting. Hence

$$\mathcal{M}_{RR}(s,\theta) = \mathcal{M}_{\rm gauge}(s,\theta)\,\mathcal{V}(z,x), \qquad \mathcal{V}(z,x) = \frac{\int dy\,\rho(y)\,V_{zy^2}(x)}{\int dy\,\rho(y)}, \tag{103}$$

$$V_w(x) \equiv \frac{\Gamma(1+wx)\,\Gamma[1+w(1-x)]}{\Gamma(1+w)}, \tag{104}$$

where $\mathcal{M}_{\rm gauge}$ carries the low-energy four-dimensional coupling and $V_w(x)$ is Eq. (38) at $\alpha' s = w$. In the same-sign channel both local invariants $\tilde{t}, \tilde{u}$ are spacelike, so the local amplitude has no poles along the radial integral. In channels containing an s ordering the local continuum approximation would smear the discrete Regge spectrum; Eq. (103) is not used there.

15.2 Radial weight and twist

For profiles $\psi_i \to (r/r_{\min})^{-\Delta_i}$ at large r, PS find that the energy dependence is governed by the twists $\tau_i = \Delta_i - \sigma_i$. The measure and profiles give $dr\, r^{3-\sum\tau_i} \propto y^{\sum\tau_i-5}dy$, i.e.

$$\rho(y) \propto y^{2\nu-1}, \qquad 2\nu = \sum_{i=1}^{4}(\tau_i - 1) \equiv \sum_i \gamma_i, \tag{105}$$

where γ_i is the twist of state i above the free spin-$\frac{1}{2}$ value. Unitarity requires $\Delta \geq 3/2$ for a spin-$\frac{1}{2}$ operator, with equality only for a free field [5]; hence $\nu \geq 0$ for normalizable states, and $\nu = 0$ only if all four are free. A weight with $\nu < 0$ is UV dominated and needs a UV boundary at y_0; this describes states with an elementary component (Sec. 15.4).

15.3 Mellin form and the disk modulus

For $z \gg 1$ only $y \lesssim z^{-1/2} \ll 1$ contributes, where the profiles are pure powers. With $w = zy^2$,

$$\mathcal{V}(z,x) \xrightarrow{z\gg1} K\,\nu\,\mathcal{M}(\nu;x)\,z^{-\nu}, \qquad \mathcal{M}(\nu;x) \equiv \int_0^\infty dw\, w^{\nu-1}\, V_w(x), \tag{106}$$

where K depends on the profiles near $r_{\min}$ ($K = 1$ for a hard wall with pure-power profiles). The exponent ν reproduces the PS counting $\mathcal{M} \sim s^{2-\sum\tau_i/2}$. The angular dependence $\mathcal{M}(\nu;x)$ does not depend on the infrared region.

Using $V_w(x) = w\,x(1-x)\,B(wx, w(1-x))$ and the Euler integral for B, the w integration can be done first:

$$\mathcal{M}(\nu;x) = \Gamma(\nu+1)\,x(1-x)\int_0^1 \frac{d\xi}{\xi(1-\xi)}\left[\ell_x(\xi)\right]^{-\nu-1}, \qquad \ell_x(\xi) = -x\ln\xi - (1-x)\ln(1-\xi). \tag{107}$$

Here ξ is the disk modulus of Eq. (13) and $e^{-w\ell_x(\xi)}$ its Koba–Nielsen factor. The radial average replaces that exponential by the power $\ell_x^{-\nu-1}$; its minimum $\ell_x(x) = H(x)$ [Eq. (40)] is the saddle that produced the flat-space suppression e^{-zH}. For a hard wall the same steps give the result at all z,

$$\mathcal{V}_{\rm HW}(z,x) = \nu z^{-\nu}\int_0^z dw\, w^{\nu-1}V_w(x) = \nu z^{-\nu}x(1-x)\int_0^1 \frac{d\xi}{\xi(1-\xi)}\,\ell_x^{-\nu-1}\,\gamma(\nu+1, z\,\ell_x)\,, \tag{108}$$

with γ the lower incomplete gamma function, and

$$\mathcal{V}_{\rm HW}(z,x) = 1 - \zeta(2)\,\frac{\nu}{\nu+2}\,z^2x(1-x) + O(z^3) \qquad (z \ll 1). \tag{109}$$

The coefficient $\nu/(\nu+2)$ in Eq. (109) depends on the infrared model; Eq. (106) does not, apart from K.

Equation (106) refines the PS counting, which fixes the exponent; the same counting was applied to open-string meson amplitudes in Ref. [3] without evaluating the angular function. In a conformal background the analogous radial integral gives a Borel-type transform of the flat amplitude [4]; the confining wall replaces it by the Mellin transform (106).

$\mathcal{M}(\nu;x)$ is meromorphic in ν. Writing $\mathcal{M} = \nu^{-1} + \int_0^1 dw\, w^{\nu-1}(V_w - 1) + \int_1^\infty dw\, w^{\nu-1}V_w$ continues it to $\mathrm{Re}\,\nu > -2$; the Taylor series of $V_w - 1$ [Eq. (29)] gives poles at $\nu = 0, -2, -3, \ldots$ with residues $1,\ -\zeta(2)x(1-x),\ \zeta(3)x(1-x), \ldots$

15.4 Regimes

Composite-dominated, $\nu > 0$. The amplitude falls as $z^{-\nu}$, a power law in place of the flat $e^{-zH(x)}$. At fixed $s \gg M_{\rm had}^2$ the angular distribution relative to the point-fermion one is $[\mathcal{M}(\nu;x)/\mathcal{M}(\nu;\frac{1}{2})]^2$, independent of s and of the infrared region.

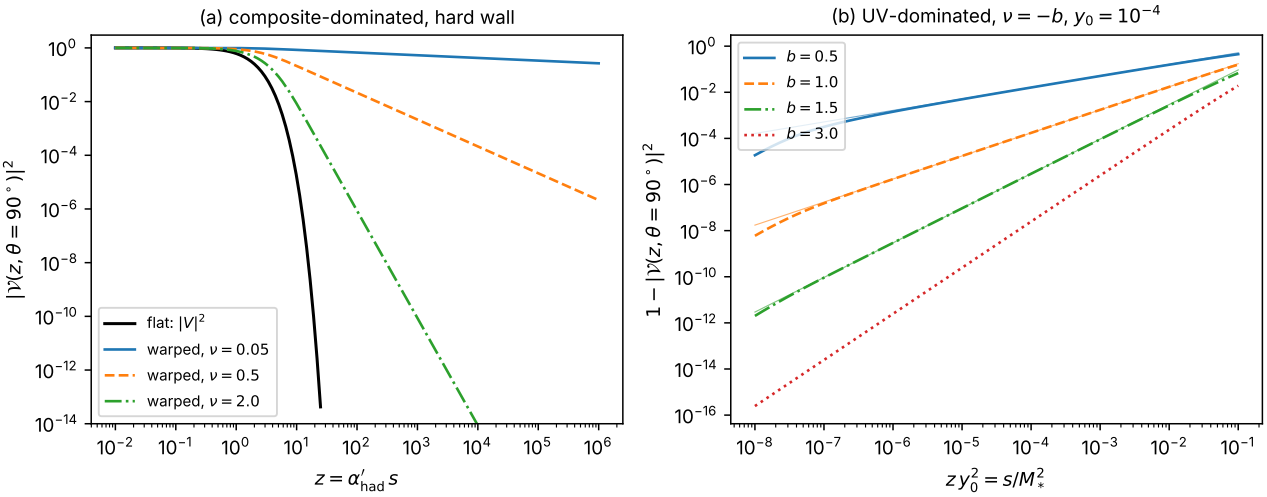


Figure 1: (a) $|\mathcal{V}|^2$ at $\theta = 90^\circ$ for the hard wall, Eq. (108), with $\nu = 0.05$, 0.5, 2, and the flat-space $|V|^2$ of Eq. (26). (b) $1 - |\mathcal{V}|^2$ for UV-dominated weights, $\nu = -b$, $y_0 = 10^{-4}$; thin lines are the leading term of Eq. (111).

Near-free, $\nu \to 0^+$.

$$\nu\, \mathcal{M}(\nu; x) = 1 + \nu\, \kappa(x) + O(\nu^2), \qquad \kappa(x) = \int_0^1 \frac{dw}{w}\, (V_w - 1) + \int_1^\infty \frac{dw}{w}\, V_w, \tag{110}$$

so $\mathcal{V} \simeq \left[z\, e^{-\kappa(x)}\right]^{-\nu}$. The hadronic slope enters only through $\nu \ln(\alpha'_{\rm had} s)$. Values:

$\chi = (1-x)/x$	1	2.33	4	9	16	99
$\kappa(x)$	0.599	0.711	0.887	1.271	1.606	2.870

For $x \to 0$, $\kappa(x) = -\ln[x \ln(1/x)] + O(1)$, so near the forward direction the radial cutoff is set by $\alpha'_{\rm had}|t| \ln(s/|t|)$, the Regge combination.

UV-dominated, $\nu = -b < 0$. With the weight (105) on $(y_0, 1]$ and $M_* \equiv M_{\rm had}/y_0$,

$$1 - \mathcal{V} = \begin{cases} b\, \big|\mathcal{M}(-b; x)\big|\, (s/M_*^2)^b, & 0 < b < 2, \\ \zeta(2)\, x(1-x)\, \dfrac{b}{b-2}\, (s/M_*^2)^2, & b > 2, \end{cases} \tag{111}$$

for $z \gg 1$ and $s \ll M_*^2$. The hard scale is M_*, set by the UV boundary rather than by $M_{\rm had}$. For $b < 2$ the leading deviation is a non-integer power of s and does not correspond to a local operator; $b = 1$ has the energy dependence of a dimension-six term with the angular function $|\mathcal{M}(-1; x)|$. At $\theta = 90^\circ$, $b|\mathcal{M}(-b; \frac{1}{2})| = 0.818,\ 0.867,\ 1.475$ for $b = 0.5,\ 1,\ 1.5$. For $b > 2$ one recovers the dimension-eight form of Eq. (31) with $M_s^4 \to M_*^4 (b-2)/b$.

Figure 1 compares the flat factor with Eqs. (108) and (111).

15.5 Consequences for the bootstrap

Constituent twist. If the two antiquarks of $[\bar{q}\bar{q}]$ are hard constituents, $\tau_i = 2$ and $\nu = 2$: $d\sigma/dt \propto s^{-6}$, the counting rule for eight constituents. With $\alpha'_{\rm had} = 0.9$ GeV^{-2} the hard-wall factor at 90° is

$|\mathcal{V}|^2 = 1.3 \times 10^{-6}$ at $\sqrt{s} = 10$ GeV (flat: 9×10^{-53}) and 1.3×10^{-22} at 1 TeV. Warping removes the exponential, but this case is excluded by the s^{-2} scaling of partonic cross sections.

Near-twist-one states. $|\mathcal{V}|^2 \propto (\alpha'_{\rm had} s)^{-\sum \gamma_i}$. Requiring the ratio to perturbative QCD to stay within 10% between $\sqrt{s} = 0.1$ and 3 TeV gives $\sum_i \gamma_i \lesssim 1.5 \times 10^{-2}$ (between 0.1 and 8 TeV, 1.1×10^{-2}). This is an order-of-magnitude estimate: it treats the extra power as multiplying the QCD prediction channel by channel. At fixed s the angular shape changes by $\sum_i \gamma_i [\kappa(x) - \kappa(\frac{1}{2})]$; between $\chi = 16$ and $\chi = 1$ this is $1.01 \sum_i \gamma_i$, the same in every mass bin.

What the warped embedding requires. Pointlike hard scattering with a hadronic Regge slope is obtained in two cases: $\nu \to 0^+$, where by the unitarity bound the state approaches a free twist-one field, or $\nu < 0$, where the state has an elementary component and the hard scale is M_*. In both cases the degree of freedom that scatters at large s is a single twist-one field. The anti-diquark–string configuration can be at most the infrared, composite component of the quark, as in partial compositeness [6]. This is consistent with the absence of partonic behavior in strongly coupled holographic duals [2].

Color. Normalizable bulk states of a holographic dual are singlets of the confining gauge group, and Chan–Paton labels on branes extending along r are global symmetries of that sector. Carrying QCD color on the Ramond strings therefore requires a confining sector in which $SU(3)_c$ is a weakly gauged global symmetry. In that construction the throat string is not the QCD flux tube, $M_{\rm had}$ is replaced by the infrared string scale of that sector (a few times the lightest Kaluza–Klein mass in Randall–Sundrum realizations [7]), and the relation $\bar{3} \wedge \bar{3} = 3$ of Eq. (56) applies to its constituents rather than to QCD antiquarks. The literal identification of Eq. (63) with the QCD string is not realized by warping.

16 Lepton–meson realization

The anti-diquark construction of Sec. VIII has a colour-singlet counterpart: the two endpoint labels form a quark–antiquark pair and the Ramond string supplies spin $\frac{1}{2}$ and lepton number. This section gives its quantum numbers and Chan–Paton structure, the string factors for the leptonic channels, and a comparison of both realizations with compositeness limits.

16.1 Quantum numbers

For each left-handed Weyl field f of one generation we list the Lorentz-scalar bilinears $\psi\chi$ or $(\psi\chi)^*$ of Standard Model fields that carry exactly the $SU(3)_c \times SU(2)_L \times U(1)_Y$ charges of f (Table 1). When $\psi = \chi$, Fermi statistics and the antisymmetric Lorentz contraction fix the flavour symmetry to equal the gauge symmetry type.

Table 1: Scalar bilinears with the gauge charges of each Standard Model field (one generation, left-handed fields). ΔB, ΔL are f minus the bilinear.

f	bilinear	ΔB	ΔL	type
u^c	$(d^c d^c)^*$, flavour antisymmetric	-1	0	anti-diquark, Eq. (58)
d^c	$(u^c d^c)^*$; QQ, flavour symmetric	-1	0	(anti-)diquark
d^c	$(QL)^*$; $u^c e^c$	0	1	leptoquark
Q	$(d^c L)^*$	0	1	leptoquark
L	$Q\, u^c$; $(Q\, d^c)^*$	0	1	meson
L	$(L\, e^c)^*$	0	1	lepton–antilepton
e^c	$(LL)^*$, flavour antisymmetric	0	1	lepton–lepton
ν^c	none			

In every entry $\Delta(B - L) = -1$. With $X = 5(B - L) - 4Y$, the left-handed fields have $X = 1$

(Q, u^c, e^c), $X = -3$ (d^c, L) and $X = 5$ (ν^c); matching Y and the non-Abelian charges forces $\Delta X = -5$, i.e. one unit of $B - L$. The string state R therefore carries the Standard Model gauge charges (none) and the $B - L$ of a right-handed neutrino. The two realizations differ in how that unit is split: $\Delta B = 1$ for $q \simeq [\bar{q}\bar{q}] \otimes R$, $\Delta L = 1$ for $\ell \simeq (q\bar{q}') \otimes R$. With lepton number conserved, the leptonic R has exactly the quantum numbers of ν_R; the baryonic R of Sec. IX has no Standard Model counterpart.

The lepton–meson identification reads

$$L_L \simeq (\bar{u}_R\, Q_L) \otimes R: \qquad \nu_L \leftrightarrow (\bar{u}_R u_L), \qquad e_L \leftrightarrow (\bar{u}_R d_L), \tag{112}$$

i.e. the u column of the chiral matrix $\Sigma_{ij} = \bar{q}_{Rj} q_{Li}$, with the charges of (π^0, π^-). The equivalent form $L \simeq (Q\, d^c)^*$ uses the d column. The singlets e_R and ν_R have no scalar partner (for e_R only the vector $\bar{u}_R \gamma^\mu d_R$). The pattern is complementary to the quark case: there u_R and d_R have scalar partners and Q_L has none. The colour singlet arises from $3 \otimes \bar{3} \supset 1$ without a junction, so none of the baryon-number questions of Sec. IX arise.

In $SU(5)$ language the entries of Table 1 are three contractions: $10 \simeq (\bar{5} \wedge \bar{5})^*$ $(u^c,\ Q,\ e^c)$; $\bar{5} \simeq \epsilon_{abcde}\, 10^{bc} 10^{de}$ $(d^c \simeq QQ,\ u^c e^c;\ L \simeq Q u^c)$; and $\bar{5} \simeq (10 \otimes \bar{5})^*$ $(d^c,\ L)$. These are the contractions of the Yukawa terms $10\, 10\, 5_H$ and $10\, \bar{5}\, \bar{5}_H$ with the Higgs replaced by a fermion bilinear. The quark–diquark relation is the plain antisymmetric product of Eq. (64); the lepton–meson relation $L \simeq Q u^c$ uses the $SU(5)$ ϵ tensor. Under $SU(3)_c$ alone the roles are reversed.

16.2 Chan–Paton realization and selection rules

The lepton–meson state is an oriented open string with labels in $N \otimes \bar{N}$. The charged Ramond state of Sec. III B, $\lambda_\pm = E_{12}, E_{21}$, is of this type: the Møller computation of Secs. III–IV is already a lepton–meson calculation. The quark–diquark state is unoriented, with labels in $\bar{N} \wedge \bar{N}$ or $\bar{N} \vee \bar{N}$ from an orientifold. If the labels carry colour only, $\Lambda^2(\bar{3}) = 3$. If they carry colour and flavour, $U(9) \supset SU(3)_c \times SU(3)_f$, then $\Lambda^2(\bar{9}) = (3, \bar{6}) \oplus (\bar{6}, 3)$ and $S^2(\bar{9}) = (3, 3) \oplus (\bar{6}, \bar{6})$: the anti-diquark that is antisymmetric in colour and in flavour requires the symmetric projection and comes with colour-sextet partners.

The oriented sector has an analogous feature: if the endpoint labels carry colour, $3 \otimes \bar{3} = 1 \oplus 8$ and the lepton has colour-octet partners, degenerate at tree level. LHC recasts exclude colour-octet leptons below 1.2–1.55 TeV [10].

For four Ramond states, the orderings with a nonzero Chan–Paton trace are those in which particle and antiparticle labels alternate. An explicit enumeration gives the same result for oriented adjoint, bifundamental, Λ^2 and S^2 labels:

channel	lepton sector	quark sector
same sign, $V(t, u)$	$e^- e^- \to e^- e^-$	$qq \to qq$
particle–antiparticle, $V(s, t)$	$e^+ e^- \to e^+ e^-$	$q\bar{q} \to q\bar{q}$
annihilation into a pair sharing one label, $V(s, t)$	$e^+ e^- \to \mu^+ \mu^-$	$q\bar{q} \to \ell^+ \ell^-$

In the last row a single ordering contributes. Its second invariant is t (between the incoming and outgoing fermions of equal charge) if the shared label has the same orientation in both states, and u otherwise; the sign of δA_{FB} below follows that orientation. If the two pairs share no label the disk amplitude vanishes; the process then needs $U(1)$ mixing beyond the disk. This makes the generation map observable: with $e \leftrightarrow d\bar{u}$, $\mu \leftrightarrow s\bar{c}$ the disk gives no $e^+ e^- \to \mu^+ \mu^-$ string factor, while with $e \leftrightarrow d\bar{u}$, $\mu \leftrightarrow s\bar{u}$ it gives $V(s, t)$. In both realizations the off-diagonal Chan–Paton gauge bosons (e.g. E_{sd}) mediate flavour-changing transitions among states that share labels and must be heavy.

Table 2: Lepton and quark sectors: published dimension-six contact-interaction limits ($\Lambda^{\pm}_{LL}$, 95% CL) and limits relevant to the same-stack string, whose leading effect is dimension eight. Entries marked "est." use Eq. (116) or a coupling mapping and are not fits.

process	dimension six, Λ^+/Λ^- [TeV]	string-sector limit	source
$ee \to ee$ (LEP)	8.3/10.3	$M_s > 0.631$ TeV (fit)	PDG [32]; Ref. [29]
$ee \to \mu\mu,\ \tau\tau$ (LEP)	8.5/9.5; 7.9/7.2	$M_s \gtrsim 0.5$–0.6 TeV (est., shared label)	PDG [32]
$q\bar{q} \to ee,\ \mu\mu$ (LHC)	24/37; 23.3/40.0	$M_s \approx 2$–3.5 TeV (est., label chain)	PDG [32]; [9]
dim.-8 dileptons (LHC)	—	ADD $\Lambda_T > 7.5$ TeV	[9]
excited lepton (first Ramond level)	—	$e^* > 5.6$ TeV (contact production)	PDG [32]
colour-octet partner	—	$\ell_8 > 1.2$–1.55 TeV (recasts)	[10]
$(g-2)_\mu$	$\Lambda \gtrsim 2.6$ (est., $\delta a \sim m_\mu^2/\Lambda^2$)	$M_s \gtrsim 0.13$ TeV (one loop, est.)	[11, 12]
$qq \to qq$ (dijet χ)	17/37	string resonances > 7.7 TeV	[33]; [31]
excited quark	—	$q^* > 6.7$ TeV	PDG [32]

16.3 Observables

From Eq. (29),

$$|V(t,u)|^2 - 1 = -\tfrac{\pi^2}{3}\, z^2 x(1-x) + O(z^3) \qquad \text{(same sign)}, \tag{113}$$

$$|V(s,t)|^2 - 1 = +\tfrac{\pi^2}{3}\, z^2 x + O(z^3) \qquad \text{(particle–antiparticle, annihilation)}. \tag{114}$$

The annihilation channel has an excess. Weighted with $1 + \cos^2\theta$ its angular average is $+\frac{\pi^2}{6} z^2$, and it shifts the forward–backward asymmetry of s-channel photon exchange, for the orientation with $V(s,t)$, by

$$\delta A_{FB} = -\frac{3\pi^2}{32}\, z^2. \tag{115}$$

A form factor that depends only on s does not produce this shift; it comes from the t dependence of V. At $\theta = 90^\circ$ a measurement of fractional precision ϵ reaches

$$M_s \gtrsim \sqrt{s}\left(\frac{\pi^2}{12\epsilon}\right)^{1/4} \text{(same sign)}, \qquad M_s \gtrsim \sqrt{s}\left(\frac{\pi^2}{6\epsilon}\right)^{1/4} \text{(annihilation, Bhabha)}. \tag{116}$$

For LEP2 Bhabha scattering, $\sqrt{s} \simeq 0.2$ TeV and $\epsilon \simeq 1.5\%$ give 0.65 TeV, in line with the fit of Ref. [29], $M_s > 0.631$ TeV, which used the model of Ref. [8], i.e. this oriented sector.

16.4 Comparison with compositeness limits

Table 2 collects the numbers. Three points follow.

(i) The string-sector limits on the scale lie a factor 5 (quarks) to 16 (leptons) below the dimension-six limits, because the leading oscillator effect is dimension eight. In length units the same-stack lepton string may be as large as $1/M_s = 3.1 \times 10^{-19}$ m against 1.9×10^{-20} m for a generic dimension-six composite electron; for quarks the corresponding values are 2.6×10^{-20} m and 5.3×10^{-21} m.

(ii) The lepton string limit, 0.63 TeV, is twelve times weaker than the quark one, 7.7 TeV, while the dimension-six limits differ by a factor 3–4. A dimension-eight effect grows as s^2, and the lepton limit comes from $\sqrt{s} = 0.2$ TeV. Future lepton colliders change this. With assumed precisions, Eq. (116) gives 2.3 TeV for FCC-ee at 365 GeV ($\epsilon = 10^{-3}$), 4.8 TeV for a 1 TeV linear collider (3×10^{-3}), 10.7 TeV for CLIC at 3 TeV (10^{-2}) and 36 TeV for a 10 TeV muon collider (10^{-2}). Bhabha scattering and δA_{FB} are the observables that distinguish the string factor from a dimension-six contact term.

(iii) Drell–Yan constrains the string only if quark and lepton labels form a single Chan–Paton chain. In the combined bootstrap, $u \simeq [\bar{d}\bar{s}]$ and $e \simeq (d\bar{u})$ share the label d, so $u\bar{u} \to e^+e^-$ has a disk contribution. Its estimated reach, 2–3.5 TeV, comes from $|V|^2 - 1 = \frac{\pi^2}{6} z^2$ at $m_{\ell\ell} \simeq 2$–3 TeV, and from mapping the ADD limit with the string amplitude lacking the $1/\alpha$ enhancement of graviton exchange.

16.5 Lepton–meson with the QCD string

Identifying the endpoint labels with QCD colour fails for three independent reasons: Bhabha scattering requires $M_s > 0.63$ TeV, against $M_{\rm had} \simeq 1$ GeV; the colour-octet partners would be degenerate with the leptons; and a hadron-sized electron gives $\delta a_e \sim (m_e/1\ \text{GeV})^2 \approx 3 \times 10^{-7}$, against $|\delta a_e| \lesssim 10^{-12}$. As for quarks (Sec. 15), the labels must belong to a separate confining sector, for which the warped analysis applies unchanged, since the same factors V appear.

If the leptonic R is lighter than the pion, Eq. (112) implies the operator $(c/\Lambda^2)(\bar{u}_R d_L)(\bar{e}_L N_R)$, whose pseudoscalar matrix element lifts the helicity suppression of $\pi \to e\nu$. The PIENU measurement of $\Gamma(\pi \to e\nu)/\Gamma(\pi \to \mu\nu)$ to 2.4×10^{-3} [13] then gives $\Lambda \gtrsim 60$ TeV for $c = 1$ (estimate). This test has no analogue in the quark sector.

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